Synthetic Maintenance Manual — DEMO-TR-01 Power Transformer

SYNTHETIC DEMONSTRATION DOCUMENT — fictional data, not a real standard

Document DEMO-TM-001 Rev B. Equipment: three-phase two-winding oil-immersed transformer, model designation DEMO-TR-01, 33 kV class of artificial design. Issued for the offline demo corpus of the Engineering RAG Assistant.

Type designation	DEMO-TR-01 / ONAF 40 MVA
Rated power (ONAF)	40 MVA (three-phase)
Rated power (ONAN)	30 MVA
Rated voltage	132/33 kV, ratio 4:1
System frequency	50 Hz
Vector group	YNd1
Tap changer	OLTC on 132 kV side, ±8 × 1.25 %

This manual is structured as: Chapter 1 Ratings, Chapter 2 Losses and efficiency, Chapter 3 Insulation and tests, Chapter 4 Protection and cooling, Chapter 5 Maintenance procedures.

Chapter 1 — Ratings

1.1 Nameplate data

The rated quantities below are nameplate values valid at rated frequency and rated voltage unless a specific operating condition is stated.

Quantity	Symbol	Value	Condition
Rated apparent power	S	40 MVA	ONAF cooling
Rated power ONAN	S	30 MVA	self-cooled
Rated high-voltage	U1	132 kV	phase-to-phase
Rated low-voltage	U2	33 kV	phase-to-phase
Rated current HV side	I1	175.0 A	at 40 MVA
Rated current LV side	I2	699.8 A	at 40 MVA
Maximum system voltage	Um	145.5 kV	HV side
Insulation level HV	LI / Pi	650 kV BIL / 275 kV	lightning / power freq
Insulation level LV	LI / Pi	170 kV BIL / 70 kV	lightning / power freq

1.2 Impedance and tolerance

Measured at rated tap and reference temperature 75 °C. Impedance tolerance is ±7.5 % of the declared value (IEC-style, synthetic).

Winding pair	Z at 75 °C	R at 75 °C	X at 75 °C
HV–LV (132/33 kV)	12.15 %	0.45 %	12.14 %

Chapter 2 — Losses and efficiency

Loss values apply to the 40 MVA ONAF rating at 75 °C reference temperature and rated voltage.

Loss item	Value	Conditions
No-load loss P0	14.2 kW	at rated voltage, 40 °C ambient
Load loss Pk	205 kW	at 75 °C, rated current
Total loss at 75 % load	130.6 kW	75 % load, 75 °C
Efficiency at 75 % load	97.88 %	40 MVA base

Worked check: P0 + 0.75^2 × Pk = 14.2 kW + 0.5625 × 205 kW = 129.6 kW; the tabulated 130.6 kW includes the stray-loss allowance of 1.0 kW.

2.1 Temperature rise limits

Ambient temperature maximum is 40 °C. Top-oil rise limit is 55 K (average winding rise limit 65 K, measured method: thermometer). At 30 MVA ONAN the average winding rise must not exceed 52 K.

Typical measured average winding rise at 40 MVA ONAF was 58.4 K, within the 65 K limit.

Chapter 3 — Insulation and routine tests

3.1 Routine power-frequency withstand test

During routine testing the 33 kV LV winding is subjected to a power-frequency test voltage of 70 kV for 1 minute. The applied voltage is phase-to-earth; for phase-to-neutral operation a value of 40.4 kV may be used on auxiliary windings only.

3.2 Insulation resistance

Insulation resistance measured with 5000 V DC shall be at least 5000 MΩ at 20 °C referred to 20 °C. Polarization index shall be at least 2.0 for windings rated above 33 kV class.

3.3 Oil test limits

Dielectric breakdown voltage of insulating oil before energization must be at least 60 kV with a 2.5 mm gap; moisture content not more than 15 mg/kg.

Chapter 4 — Protection and cooling

4.1 Bookholz and pressure protection

The transformer is protected by a 630 kPa setting pressure-relief vacuum device, sudden-pressure relay set at 0.65 bar, and Buchholz relay with gas volumes 150–300 cm³ (first stage) and 300 cm³ (second stage).

4.2 Differential protection

The 87T differential relay restraining threshold is 0.25 × I_rated (52.5 A at 209.5 A LV rated current) with a second-harmonic restraint blocking criterion at 15 %.

4.3 Cooling fans

Four forced-cooling fans of 0.75 kW each; total fan motor load is 3.0 kW. Fan contactor duty: continuous at 40 °C ambient. Fans start automatically at top-oil temperature 65 °C and stop at 55 °C.

Chapter 5 — Maintenance and safety

5.1 Isolation procedure

SAFETY: De-energize all windings, apply the 240 mm² temporary earthing set (three-phase short-circuited and earthed), verify absence of voltage phase-by-phase, and tag before entry. The temporary earth conductor shall not be smaller than 25 mm² copper.

5.2 Oil sampling

Take the sample from the bottom valve with at least 5 times the valve volume flushed. For dielectric testing, allow gas to equilibrate 15 min before the breakdown test.

5.3 Re-tightening and torque

Bushing terminal torque: M12 bolts 45 Nm, M16 bolts 110 Nm. Torque shall be applied at ambient temperature 20 to 40 °C.

Any figure in this synthetic manual is fictional; a real project must use the actual test report and OEM manual.